1. Cleaning Messy Data

Code flow:-

- Import Pandas as pd import Seaborn as sns, these are the tools.
- Pandas gives information about Tables and Seaborn gives us ready made datasets.
- we have to load the dataset into table called data.
- Data must captain useful columns called Survived, Pclass, sex, age, fare, embarked.
- Check and fill missing values using median.
- convert categorical values into numerical values using get dummies()
- Display the cleaned dataset.

2. Train & Test fairly (KNN on Iris Dataset).

Code flow:-

- Import the required libraries
- load the iris dataset
- split the dataset into training and testing.
- test size=0.2 means 80% training and 20% testing data.
- If changed to 0.3, it becomes 70% training & 30% testing
- Apply KNN algorithm.
- Calculate the display the accuracy.

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3. Predict A Number (Linear Regression on Auto MPG Dataset)

Code flow:-

- Import Pandas, Numpy and Scikit-learn.
- load to Auto MPG dataset.
- Select input and output features.
- Train the Linear Regression model.
- fit() is used to train the model.
- If removed the model will not be trained.
- Predict mileage and display the result.

4. Predict Yes/NO (KNN & SVM on Social Networking)

Codeflow:-

- Import the required libraries.
- load the Social Network Ads dataset.
- Separate input and output features.
- Apply K-Means Clustering.
- Predict() is used to predict the output.
- It is called after training the model.
- Apply KNN & SVM algorithms and display accuracy.

5. And Groups (K-Means on Mall Customers).

Codeflow:-

- Import Pandas, matplotlib & K-means.
- load to Mall Customers dataset.
- Select annual incomes and spending score.
- Apply K-means clustering.
- accuracy_score() is used to calculate model accuracy
- Higher accuracy means more correct predictions.

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6. Groups as a Tree (Hierarchical clustering on mall customers)

Code flow:-

- Import the required libraries
- load the mall customers dataset.
- select the required features.
- Apply Hierarchical clustering
- Draw and display the dendrogram.

7. A Forest of Trees (Random forest on Breast Cancer)

Code flow:-

- Import the required libraries.
- load the Breast Cancer dataset.
- split the data into training and testing sets
- Apply the Random Forest algorithm.
- predict the result and calculate accuracy.

8. Boosting (Gradient Boosting on Heart Diseases)

code flow:-

- Import the required libraries
- load the heart Diseases dataset
- Separate input and output features.
- Train the Boosting model.
- predict the result and display accuracy

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